



MAKERERE UNIVERSITY

COLLEGE OF COMPUTING AND INFORMATION SYSTEM
BSE4104 EMERGING TRENDS IN SOFTWARE ENGINEERING

WEEK 4: HUMAN IN LOOP AUTHORIZATION POLICY

PROJECT: University Student-Support Case Agent

STUDENT DETAILS:

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1. Executive Summary & Purpose

Policy basis. The BSE4104 Capstone Brief (Sections 2 & 4) specifies:

“Keep high-impact actions behind human approval... Implement input/output validation, tool allow-list, authorization, limits and approval controls.”

This policy establishes the operational boundaries between autonomous agent execution, mandatory human approval gates, and strictly prohibited actions for the University Student-Support Case Agent.

2. Action Classification Matrix

Every tool call and user intent is categorized into one of three risk tiers. The classification determines whether the agent may execute directly, must obtain staff approval, or must reject the request and refer the student to the appropriate human office.

Student Request	Intent Analysis	Risk Tier	System Action	Outcome
Admissions / Grades / Fees / Disciplinary	Intent identified as prohibited	Tier 3	Reject + refer	Deterministic rejection
Status / routine support	Low-impact intent	Tier 1	Execute approved tool	Immediate response / OPEN ticket
Exam / urgent support	Moderate or high impact	Tier 2	Approval gate	PENDING_STAFF_APPROVAL

Tier 1: Autonomous Execution (Low Impact)

Actions that do not create binding legal, financial, or academic commitments:

- Procedural Q&A: Answering handbook questions grounded strictly on retrieved chunks from the verified corpus.
- Case Status Retrieval (get_case_status): Deterministic database queries that output current state without modifying records.
- Routine Support Ticket Creation (create_support_ticket): Tickets classified under GENERAL_QUERY or TIMETABLE with urgency LOW or MEDIUM.
- Routine tickets are immediately assigned status OPEN and queued for routine staff review.

Tier 2: Human-in-the-Loop Approval Gate (Moderate to High Impact)

Actions where an automated outcome could adversely affect a student’s academic standing or departmental operations:

- Examination Queries & Disputes (category == EXAMINATION): missing marks, examination clashes, or special examination requests.
- Enforcement: the tool sets status to PENDING_STAFF_APPROVAL and requires_human_approval to true.
- Student communication: the agent states that the ticket has been logged and flagged for priority review by department staff before action is taken.
- High-Urgency Escalations (urgency == HIGH): severe academic disruption or looming deadlines.
- High-urgency cases remain locked until designated support staff or the course coordinator signs off in the triage queue.

Tier 3: Prohibited Actions (Strict Out-of-Scope)

Actions that the agent is architecturally prevented from performing under any circumstances:

- Modifying grade records or examination results. The agent has no tool to alter marks, approve retake waivers, or recalculate CGPA.
- Financial and tuition fee waivers. The agent cannot check or alter bank accounts, tuition balances, or disburse scholarships.
- Official disciplinary actions and admissions. Admissions eligibility and disciplinary committee decisions remain strictly human-run processes.
- Arbitrary execution / shell access. The agent cannot execute arbitrary code or query external unapproved networks.

3. Enforcement Mechanisms

1. **Orchestration Tool Allow-List.** The model orchestrator (src/model_client.py) only receives schema definitions for approved tools: `get_case_status` and `create_support_ticket`. The model cannot invoke any undeclared function.
2. **Deterministic Pre-Validation.** Before executing `create_support_ticket`, deterministic software inspects parameters. If an unauthorized request is detected (for example, asking to change a grade), the request is intercepted before database insertion and returns error `DISALLOWED_TOPIC`.
3. **State Locking.** Tickets with high risk are saved in the mock database with status = `PENDING_STAFF_APPROVAL`. The system prohibits transitions to `RESOLVED` or `CLOSED` without human operator intervention.

4. Student Communication Protocol for Human Approval

Required Agent Message

“I have drafted and logged your ticket (ID: [TICKET_ID]) under category [CATEGORY]. Because this involves an examination or urgent procedural issue, it has been forwarded to the Department Support Coordinator for human review. You can check the status of your case using this ticket ID.”

Prohibited Agent Statements

- “Your examination clash has been resolved.”
- “I have approved your retake registration.”
- “Your grade has been updated in the portal.”

5. Auditability and Traceability

Compliance reference: User Story US-8.

Every tool invocation, whether successful, intercepted, or flagged for approval, is written to an execution trace log in `evidence/traces/`.

Each log entry records:

- Timestamp (ISO 8601)
- Student Identifier (anonymized/synthetic)
- Tool Name & Raw Parameters
- Approval Flag (requires_human_approval)
- Execution Result / Error Code

Policy Summary at a Glance

Tier	Risk Level	Agent Authority	Required Outcome
Tier 1	Low	Autonomous execution	Answer or OPEN routine ticket
Tier 2	Moderate–High	Draft/log only	PENDING_STAFF_APPROVAL + human review
Tier 3	Prohibited	No execution	Reject + human office referral